# 1. Dataset & Exploratory Data Analysis (EDA)



**Task & Dataset Selection**:

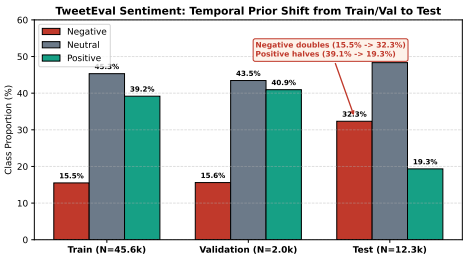
- **Benchmark**: `cardiffnlp/tweet_eval` (3-class Sentiment: Neg, Neu, Pos).
- **Split Sizes**: 45, 615 Train | 2, 000 Validation | 12, 284 Official Test.

**Critical Finding: Severe Temporal Prior Shift**:

- **Train / Val**: 15.5% Negative, 45.3% Neutral, 39.1% Positive.
- **Official Test**: **32.3% Negative** ($\times$**2.1** doubles!), 48.3% Neutral, **19.3% Positive** ($\times$**0.5** halves!).
- *Why Macro-F1*: Raw accuracy is misleading under prior shift. **Macro-F1** ($\frac{1}{3}\sum_c F_{1,c}$) weights all three classes equally.
- *Class-Weighted Loss*: Inverse frequency weights $w_c \propto 1/N_c$ prevent the majority Neutral class from overwhelming minority Negative gradients.

**Twitter Stylometrics**:

- Mean length: 19.4 words/tweet; noisy syntax, user mentions, hashtags, slang.
- **Negations**: 3$\times$ more frequent in Negative (0.224) vs. Positive (0.079).
- **Exclamations**: 3$\times$ more frequent in Positive (0.468) vs. Negative (0.157).




## Dataset & Metric Summary

Because Negative tweets double in the test set (15.5% $\rightarrow$ 32.3%), Macro-F1 and class weighting are essential for fair evaluation.

# 2. Text Segmentation: Comparing 3 Tokenizers

Topic 2: Whitespace vs. NLTK TweetTokenizer vs. Subword BPE

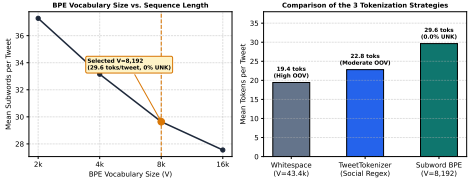
**Comparing Three Tokenization Strategies**:

1. **Whitespace** (`text.split()`): $V = 43,395$, 19.44 toks/tweet (ratio 1.00). Leaves punctuation glued to words (`"it..."`); high OOV rate on unseen tweets.
2. **NLTK** `TweetTokenizer` **(Social Regex)**: $V = 43,395$, 22.81 toks/tweet (ratio 1.17). Preserves emoticons (`:-)`), strips `@handles`, normalizes repeated chars (`coooool` → `coool`). Moderate OOV on novel typos/slang.
3. **Subword Byte-Level BPE**: Trained in-domain on training tweets ⇒ **0.00% UNK rate**. Splits rare words into reusable subword stems and affixes.

**Byte-BPE Vocabulary Sweep ($V \in \{2\mathbf{k}, 4\mathbf{k}, 8\mathbf{k}, 16\mathbf{k}\}$)**:

- $V = 2,048$: Over-fragmented into short pieces (37.28 toks/tweet, 1.92 ratio).
- **V = 8, 192 (Selected)**: 29.64 toks/tweet (1.52 ratio). Sweet spot: preserves stems while keeping sentiment affixes (`un-`, `dis-`, `'t`) with 0% UNK.
- $V = 16,384$: 27.55 toks/tweet (1.42 ratio); memorizes rare noise, doubles params.

Ex: `"cooool :-) didn't"` → **TweetTok**: `['cooool', ':-)', "didn't"]` | **BPE-8k**: `['c', 'oo', 'ool', 'Ġ:-)', 'Ġdidn', "'t"]`




## Segmentation Takeaway

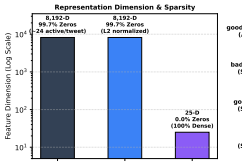
Rule-based `TweetTokenizer` normalizes social media syntax for word embeddings, while BPE ($V = 8,192$) ensures 0.00% OOV and morphological sharing.

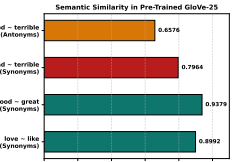
**Three Sparse Subword Spaces (BPE, $V = 8,192$, 99.71% Zeros)**:

1. **Binary Token Indicator**: $x_i \in \{0, 1\}$. Marks presence/absence. Since subwords repeat in $< 2.4\%$ of tweet positions, indicator captures $\sim 97.6\%$ of count information.
2. **Bag-of-Words (BoW) Counts**: $x_i = \text{count}(i, d) \in \mathbb{N}_0$. Preserves raw count magnitude for repeated punctuation (`!!!`) and intensifiers.
3. **Subword TF-IDF**: $x_i = \text{TF}(i, d) \cdot \text{IDF}(i)$ with $L_2$ normalization ($\|x\|_2 = 1$). On short tweets, $L_2$ normalization penalizes tweets with neutral filler words by shrinking the magnitude of strong sentiment tokens.

**Pre-Trained Dense Twitter Embeddings (`glove-twitter-25`, $D = 25$)**:

- Pre-trained on 2B tweets (1.19M word vocabulary, 0% zeros).
- Captures continuous synonymy (`cos(good, great)`=0.94, `cos(love, like)`=0.90), whereas sparse BoW treats all distinct words as orthogonal (cos = 0).
- **Mean-Pooled ($\mathbb{R}^{25}$)**: Orderless average vector (for Log-Linear & Shallow MLP).
- **Sequential ($\mathbb{R}^{T \times 25}$)**: Preserves ordered trajectory $[e(w_1), \ldots, e(w_T)]$ (for BiLSTM).




## Interpretability of Semantic Clustering

Cosine similarities in GloVe: `good` ≈ `great` (0.94), `love` ≈ `like` (0.90), `bad` ≈ `terrible` (0.80). Dense embeddings group synonyms into semantic neighborhoods.

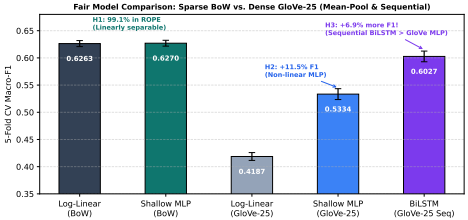
# 4. Model Architectures & Fair Hypotheses

**Three Clean Architectures Evaluated**:

1. **Log-Linear Softmax**: $\hat{y} = \text{softmax}(Wx + b)$. Direct linear decision boundary ($24,579$ params on BoW; only 78 params on 25-D GloVe).
2. **Shallow MLP**:
   $\text{Linear}(D, 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.25) \rightarrow \text{Linear}(256, 3)$.
   Single hidden layer with non-linear activation (2.10M params on BoW; $7,427$ on GloVe).
3. **Bidirectional LSTM**:
   $\text{BiLSTM}(25, 64) \rightarrow \text{MeanPool} \rightarrow \text{Linear}(128, 3)$. Sequential recurrence ($96,643$ parameters).

**Three Fair Experimental Hypotheses**:

- **H1 (Sparse BoW, 8,192-D)**: Log-Linear ($24,579$ params) vs. Shallow MLP ($2,098,051$ params). High-dimensional BoW is already linearly separable $\Rightarrow$ MLP adds no gain.
- **H2 (Dense GloVe-25 Mean-Pool)**: Log-Linear (78 params) vs. Shallow MLP ($7,427$ params). A flat plane underfits 25-D space $\Rightarrow$ non-linear ReLU yields $+11.5\%$ F1.
- **H3 (Fair MLP vs. LSTM on Dense GloVe-25)**: Shallow MLP (Mean-Pooled $\mathbb{R}^{25}$) vs. BiLSTM (Sequential $\mathbb{R}^{T \times 25}$). Exact same `TweetTokenizer` + `glove-twitter-25` embeddings $\Rightarrow$ sequential recurrence captures word order & negation!




## Architecture Complexity Matrix

| Model Architecture | Input Space | Params | Hypothesis Tested |
|---|---|---|---|
| Log-Linear | Sparse BoW (8, 192-D) | 24,579 | H1 Baseline (Linear) |
| Shallow MLP | Sparse BoW (8, 192-D) | 2,098,051 | H1 Test (Non-linear) |
| Log-Linear | GloVe Mean-Pool (25-D) | 78 | H2 Baseline (Flat plane) |
| Shallow MLP | GloVe Mean-Pool (25-D) | 7,427 | H2 Test / H3 Baseline |
| Bidirectional LSTM | GloVe Sequence ($T \times 25$) | 96,643 | H3 Fair Sequential Test |

# 5. 5-Fold Stratified CV & Correlated Bayesian ROPE Decisions

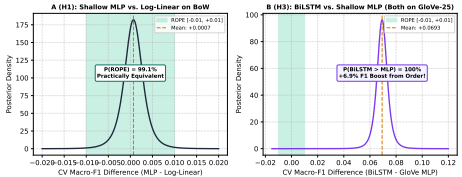
**Protocol: 5-Fold Stratified CV ($N = 45,615$) & Bayesian $t$-Test**:

- Folds share 80% training data. We use the **Correlated Bayesian $t$-test** ($\rho = 0.20$) with ROPE $= [-0.01, +0.01]$ ($\pm 1\%$ Macro-F1).

| Model Configuration | Representation | 5-Fold CV Macro-F1 | CV Accuracy |
|---|---|---|---|
| Log-Linear (Indicator) | Binary Indicator (8, 192-D) | $0.6259 \pm 0.0038$ | $65.10\% \pm 0.43\%$ |
| **Log-Linear (BoW)** | **Bag-of-Words (8, 192-D)** | $\mathbf{0.6263 \pm 0.0055}$ | $\mathbf{65.08\% \pm 0.56\%}$ |
| Log-Linear (TF-IDF) | Subword TF-IDF (8, 192-D) | $0.6058 \pm 0.0028$ | $62.89\% \pm 0.35\%$ |
| **Shallow MLP (BoW)** | **Bag-of-Words (8, 192-D)** | $\mathbf{0.6270 \pm 0.0054}$ | $\mathbf{65.13\% \pm 0.68\%}$ |
| Log-Linear (GloVe-25) | GloVe Mean-Pool (25-D) | $0.4187 \pm 0.0072$ | $52.84\% \pm 0.81\%$ |
| Shallow MLP (GloVe-25) | GloVe Mean-Pool (25-D) | $0.5334 \pm 0.0100$ | $58.74\% \pm 0.94\%$ |
| **BiLSTM (GloVe-25)** | **GloVe Sequence ($T \times 25$)** | $\mathbf{0.6027 \pm 0.0098}$ | $\mathbf{64.70\% \pm 0.99\%}$ |

**5 Fair Pairwise Correlated Bayesian Decisions**:

- **BoW vs. Indicator**: $\Delta = +0.0004 \Rightarrow \mathbf{P}(\text{ROPE}) = \mathbf{98.8}\%$ (Equivalence: words rarely repeat).
- **BoW vs. TF-IDF**: $\Delta = +0.0205 \Rightarrow \mathbf{P}(\text{BoW} > \text{TF-IDF}) = \mathbf{99.4}\%$ (BoW wins: $L_2$ norm dilutes).
- **H1 (MLP vs. Log-Lin on BoW)**: $\Delta = +0.0007 \Rightarrow \mathbf{P}(\text{ROPE}) = \mathbf{99.1}\%$ (Practically equivalent!).
- **H2 (MLP vs. Log-Lin on GloVe)**: $\Delta = +0.1147 \Rightarrow \mathbf{P}(\text{MLP} > \text{Linear}) = \mathbf{100}\%$ ($+11.5\%$ F1 jump).
- **H3 (BiLSTM vs. MLP on GloVe)**: $\Delta = +0.0693 \Rightarrow \mathbf{P}(\text{LSTM} > \text{MLP}) = \mathbf{100}\%$ ($+6.9\%$ F1 from sequence!).




## Statistical Verification Summary

ROPE confirms Shallow MLP $\approx$ Log-Linear on BoW (99.1%), while BiLSTM decisively outperforms Shallow MLP on the same GloVe embeddings (100%).

# 6. Official Test Set ($N = 12,284$) & Error Analysis

Topic 6: Test Benchmark, Confusion Matrices & Failure Modes

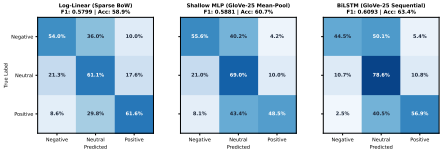
**Official Held-Out Test Set Benchmark ($N = 12,284$):**

| System | Params | Test Macro-F1 | Test Acc | Key Characteristic |
|---|---|---|---|---|
| **Log-Linear (BoW)** | **24,579** | **0.5799** | **58.91%** | Selected linear champion (from-scratch) |
| Shallow MLP (BoW) | 2,098,051 | 0.5862 | 59.46% | Confirms sparse linear equivalence |
| Shallow MLP (GloVe-25) | 7,427 | 0.5881 | 60.70% | Pre-trained dense, but orderless |
| **BiLSTM (GloVe-25)** | **96,643** | **0.6093** | **63.37%** | **Best Overall Champion (+2.12% F1)** |

**Qualitative Error Analysis Across All 3 Systems:**

| Test Tweet | True | BoW Linear | GloVe MLP | GloVe BiLSTM |
|---|---|---|---|---|
| *"An interesting security vulnerability - albeit not for the everyday car thief..."* | Neu | Pos (*interesting*) | Neg (*vulnerability*) | **Neu (Correct!)** |
| *"Christians In Palestine: Yes, They Exist #gaza #israel"* | Neu | Neg (*hashtags*) | Neg (*conflict*) | **Neu (Correct!)** |
| *"Wow, first Hugo Chavez and now Fidel Castro... running out of heroes."* | Neg | Pos (*Wow/heroes*) | Neu (*cancels out*) | Pos (Sarcasm failure) |

- **Negation Scope**: BoW and GloVe MLP miss qualifier; sequential BiLSTM resolves *albeit not for...* correctly.
- **Topic Bias**: Orderless models trigger on conflict hashtags; BiLSTM parses *Yes, They Exist* as Neutral.
- **Sarcasm & World Knowledge**: All models fail due to lack of pragmatics/irony.




## Why Sequential Embeddings Win on Test

Reading GloVe embeddings sequentially resolves negation scope and syntactic assertions, unlocking $+2.12\%$ F1 and $+2.67\%$ accuracy over mean-pooled MLP!